

Social Computing

Unit V: Social Media Behavior

Syllabus:

Recommendation in social media: Challenges, Classical Recommendation Algorithms, Recommendation using Social Context, Evaluating Recommendations, Behaviour Analytics: Individual Behaviour, Collective Behaviour

❖ Recommendation in social media

Recommendation in Social Media

Recommendation in Social Media is the process of suggesting relevant content, people, products, groups, pages, videos, or advertisements to users based on their interests, behavior, and interactions on social networking platforms.

Definition

A recommendation system analyzes user data such as likes, comments, shares, follows, search history, and browsing patterns to provide personalized suggestions that improve user engagement and experience.

Objectives of Social Media Recommendation

1. Deliver personalized content to users.
2. Increase user engagement and satisfaction.
3. Help users discover new friends, communities, and content.
4. Improve retention and time spent on the platform.
5. Support targeted advertising and marketing.

Types of Recommendations in Social Media

1. Content Recommendation

Suggests posts, videos, articles, and stories based on user interests.

Example: A user who frequently watches cooking videos is recommended more cooking-related content.

2. Friend Recommendation

Suggests people a user may know based on mutual friends, location, interests, or interactions.

Example: "People You May Know" feature.

3. Group or Community Recommendation

Suggests groups, communities, or forums that match user interests.

Example: Recommending photography groups to users interested in photography.

4. Product Recommendation

Suggests products based on user preferences and previous purchases.

Example: Showing fashion products similar to those previously viewed.

5. Advertisement Recommendation

Displays targeted advertisements according to user demographics and behavior.

Example: Showing smartphone ads to users searching for mobile reviews.

Recommendation Techniques

1. Content-Based Filtering

- Recommends items similar to those a user liked before.
- Uses item features and user preferences.

Example: If a user likes technology articles, similar articles are recommended.

2. Collaborative Filtering

- Recommends items liked by users with similar interests.
- Based on user-user or item-item similarity.

Example: Users who liked the same movies may receive similar recommendations.

3. Hybrid Recommendation

- Combines content-based and collaborative filtering.
- Provides more accurate recommendations.

4. Graph-Based Recommendation

- Uses social network relationships and interactions.
- Explores connections among users and content.

Recommendation Process

1. Data Collection

- a. Gather user profiles, activities, likes, shares, and comments.

2. User Modeling

- a. Create a profile representing user interests.

3. Similarity Calculation

- a. Find similar users or content.

4. Recommendation Generation

- a. Generate a ranked list of recommendations.

5. Feedback Collection

- a. Use user reactions to improve future recommendations.

Applications

- News Feed Personalization
- Friend Suggestions
- Video Recommendations
- Online Shopping Recommendations
- Targeted Advertising
- Community Discovery

Advantages

- Improves user experience.
- Saves time in finding relevant content.
- Increases user engagement.
- Helps discover new interests and connections.

Limitations

- Privacy concerns due to extensive data collection.
- Cold-start problem for new users.
- Filter bubble effect (limited exposure to diverse content).
- Data sparsity issues.

Conclusion

Recommendation systems are a fundamental component of social media platforms. They use user behavior, interests, and social connections to provide personalized suggestions, enhancing user engagement and helping users discover relevant content, people, and communities.

❖ Challenges

Challenges in Social Media Recommendation Systems

Social media recommendation systems face several challenges while providing accurate and personalized suggestions.

1. Cold Start Problem

- Occurs when a new user or new item has little or no interaction data.
- The system finds it difficult to make accurate recommendations.

Example: A newly registered user has no likes, follows, or viewing history.

2. Data Sparsity

- Most users interact with only a small fraction of available content.
- Insufficient data makes similarity calculations difficult.

Example: A user likes only a few posts among millions of available posts.

3. Scalability

- Social media platforms contain millions of users and billions of posts.
- Recommendation algorithms must process large amounts of data efficiently.

4. Dynamic User Interests

- User preferences change over time.
- The system must continuously update user profiles.

Example: A user interested in sports today may become interested in travel later.

5. Information Overload

- Huge volumes of content are generated every second.
- Selecting the most relevant content becomes challenging.

6. Privacy and Security Issues

- Recommendation systems collect large amounts of personal data.
- Protecting user privacy while maintaining recommendation quality is difficult.

7. Filter Bubble and Echo Chamber

- Users may repeatedly receive similar content.
- Exposure to diverse opinions and information decreases.

Example: A user interested in a political topic sees only content supporting that viewpoint.

8. Fake Accounts and Spam

- Bots and fake profiles can manipulate recommendations.
- They may promote misleading content or products.

9. Trust and Transparency

- Users often do not understand why certain recommendations are shown.
- Lack of transparency can reduce trust in the platform.

10. Diversity vs. Accuracy Trade-off

- Highly accurate recommendations may lack variety.
- Diverse recommendations may reduce accuracy.

11. Context Awareness

- Recommendations should consider time, location, device, and current activity.
- Capturing context accurately is challenging.

12. Real-Time Recommendation

- Social media content changes rapidly.
- Systems must generate recommendations instantly while handling large-scale data.

Conclusion

The major challenges in social media recommendation systems include **cold start, data sparsity, scalability, changing user interests, privacy concerns, filter bubbles, spam, transparency, diversity, context awareness, and real-time processing**. Addressing these challenges is essential for delivering accurate, relevant, and trustworthy recommendations.

❖ Classical Recommendation Algorithms

Nov 2024

Q5) a) What do you mean by recommendation in social context? Explain any one recommendation algorithm.

Ans. Recommendation in Social Context

Recommendation in a social context refers to the process of suggesting relevant items such as friends, products, movies, groups, news, or content to users by utilizing information from social networks, user behavior, preferences, and relationships among users.

The main goal is to help users discover items that match their interests while leveraging social connections and interactions.

Examples

- Friend suggestions on social media platforms.
- Movie recommendations based on friends' ratings.
- Product recommendations in online shopping.

- Group or community recommendations.

Recommendation Algorithm: Collaborative Filtering

Collaborative Filtering (CF) is one of the most widely used recommendation algorithms. It recommends items to a user based on the preferences and behaviors of similar users.

Working Principle

The basic assumption is:

"Users who had similar interests in the past are likely to have similar interests in the future."

Steps of Collaborative Filtering

1. Collect User Data

- a. Gather ratings, likes, purchases, or interactions.

2. Find Similar Users

- a. Calculate similarity between users using measures such as Cosine Similarity or Pearson Correlation.

3. Identify Preferred Items

- a. Find items liked by similar users.

4. Generate Recommendations

- a. Recommend items that similar users liked but the target user has not yet interacted with.

Example

Consider the following movie ratings:

User	Movie A	Movie B	Movie C
User 1	5	4	?
User 2	5	4	5
User 3	2	1	2

- User 1 and User 2 have similar ratings.
- User 2 rated Movie C highly.
- Therefore, the system recommends **Movie C** to User 1.

Types of Collaborative Filtering

1. User-Based Collaborative Filtering

- Finds users with similar interests.

- Recommends items liked by similar users.

2. Item-Based Collaborative Filtering

- Finds items similar to those previously liked by the user.
- Recommends similar items.

Advantages

- Does not require detailed item information.
- Provides personalized recommendations.
- Effective when sufficient user data is available.

Limitations

- Cold start problem for new users/items.
- Data sparsity issues.
- Scalability challenges for large datasets.

Conclusion

Recommendation in a social context helps users discover relevant content, products, or people by analyzing social relationships and user behavior. **Collaborative Filtering** is a popular recommendation algorithm that recommends items based on the preferences of similar users, making it effective for personalized recommendations in social media and e-commerce platforms.

Q6) a) Explain evaluation measures for recommendation algorithms.

Ans. Evaluation Measures for Recommendation Algorithms

Evaluation measures are used to assess the effectiveness and quality of recommendation algorithms. They help determine how accurately and usefully a system recommends items to users.

1. Precision

Precision measures the proportion of recommended items that are actually relevant to the user.

Formula

$$Precision = \frac{\text{Relevant Recommended Items}}{\text{Total Recommended Items}}$$

Example

- Recommended items = 10

- Relevant recommended items = 8

Precision = $8/10 = 0.8$ (80%)

Significance

- High precision means most recommendations are useful.

2. Recall

Recall measures the proportion of relevant items that are successfully recommended.

Formula

$$Recall = \frac{\text{Relevant Recommended Items}}{\text{Total Relevant Items}}$$

Example

- Total relevant items = 20
- Relevant recommended items = 8

Recall = $8/20 = 0.4$ (40%)

Significance

- High recall means the system finds most relevant items.

3. F1-Score

F1-Score combines Precision and Recall into a single metric.

Formula

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Significance

- Useful when both precision and recall are important.

4. Accuracy

Accuracy measures the proportion of correct recommendations among all predictions.

Formula

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Significance

- Provides an overall performance measure.
- Less informative when data is highly imbalanced.

5. Mean Absolute Error (MAE)

Measures the average difference between predicted ratings and actual ratings.

Formula

$$MAE = \frac{1}{N} \sum |Predicted\ Rating - Actual\ Rating|$$

Significance

- Lower MAE indicates better prediction accuracy.

6. Root Mean Square Error (RMSE)

Measures prediction error while giving greater weight to larger errors.

Formula

$$RMSE = \sqrt{\frac{1}{N} \sum (Predicted\ Rating - Actual\ Rating)^2}$$

Significance

- Lower RMSE indicates more accurate recommendations.

7. Coverage

Coverage measures the percentage of items that the recommender system can recommend.

Formula

$$Coverage = \frac{Number\ of\ Recommended\ Items}{Total\ Items}$$

Significance

- Higher coverage means a wider range of items can be recommended.

8. Diversity

Measures how different the recommended items are from one another.

Significance

- Prevents users from receiving only similar recommendations.

- Improves discovery of new content.

9. Novelty

Measures whether recommended items are new or previously unknown to users.

Significance

- Encourages exploration.
- Helps users discover fresh content.

10. Serendipity

Measures how surprising and useful recommendations are.

Example

Recommending an unexpected but interesting movie that the user ends up enjoying.

Significance

- Improves user satisfaction.

Summary Table

Measure	Purpose
Precision	Fraction of recommended items that are relevant
Recall	Fraction of relevant items recommended
F1-Score	Balance between precision and recall
Accuracy	Overall correctness
MAE	Average prediction error
RMSE	Prediction error with emphasis on large errors
Coverage	Range of recommendable items
Diversity	Variety among recommendations
Novelty	Ability to suggest new items
Serendipity	Ability to provide unexpected useful recommendations

Conclusion

Evaluation measures are essential for assessing recommendation systems. **Precision, Recall, F1-Score, MAE, RMSE, Coverage, Diversity, Novelty, and Serendipity** are commonly used metrics. Together, they help determine how accurate, useful, diverse, and user-friendly a recommendation algorithm is.

Apr 2023

Q6) b) Explain in details Classical Recommendation Algorithms.

Ans. Classical Recommendation Algorithms

Classical recommendation algorithms are traditional techniques used to suggest items (movies, products, books, friends, etc.) to users based on their preferences, ratings, or behavior. These algorithms form the foundation of modern recommender systems.

1. Content-Based Filtering

Definition

Content-Based Filtering recommends items similar to those that a user has liked in the past. It uses the characteristics or features of items to make recommendations.

Working

1. Create a user profile based on previously liked items.
2. Analyze item features (keywords, categories, attributes).
3. Recommend items with similar features.

Example

- A user likes science-fiction movies.
- The system recommends other science-fiction movies.

Advantages

- Personalized recommendations.
- Does not require data from other users.

Disadvantages

- Limited diversity.
- Difficult to recommend completely new types of items.

2. Collaborative Filtering

Definition

Collaborative Filtering recommends items based on the preferences of similar users.

Principle

Users with similar interests are likely to prefer similar items.

Types

a) User-Based Collaborative Filtering

- Finds users similar to the target user.
- Recommends items liked by similar users.

b) Item-Based Collaborative Filtering

- Finds items similar to those the user previously liked.
- Recommends similar items.

Example

- User A and User B like the same movies.
- If User B likes a new movie, it is recommended to User A.

Advantages

- Provides highly personalized recommendations.
- Does not require item features.

Disadvantages

- Cold-start problem.
- Data sparsity issues.

3. Demographic-Based Recommendation

Definition

Recommendations are generated using demographic information such as age, gender, occupation, and location.

Example

- Young users may receive recommendations for trending games.
- Professionals may receive recommendations for business articles.

Advantages

- Simple implementation.
- Useful for new users.

Disadvantages

- May not capture individual preferences accurately.

4. Knowledge-Based Recommendation

Definition

Recommendations are generated using explicit knowledge about user needs and item characteristics.

Working

- User specifies requirements.
- System matches requirements with item features.

Example

- A user wants a laptop under ₹50,000 with 16 GB RAM.
- The system recommends suitable laptops.

Advantages

- No need for historical ratings.
- Suitable for infrequent purchases.

Disadvantages

- Requires detailed domain knowledge.

5. Utility-Based Recommendation

Definition

Items are recommended based on a utility function that estimates the usefulness of an item to a user.

Example

- Recommending a flight based on price, travel time, and comfort.

Advantages

- Flexible and customizable.
- Considers multiple factors.

Disadvantages

- Utility function can be difficult to design.

6. Hybrid Recommendation

Definition

Combines two or more recommendation techniques to improve performance.

Example

- Combining Content-Based Filtering and Collaborative Filtering.

Working

1. Generate recommendations using multiple methods.
2. Merge or rank results.
3. Present the final recommendations.

Advantages

- Higher accuracy.
- Reduces cold-start and sparsity problems.

Disadvantages

- More complex implementation.

Comparison of Classical Recommendation Algorithms

Algorithm	Basis	Advantages	Limitations
Content-Based Filtering	Item features	Personalized	Limited diversity
Collaborative Filtering	Similar users/items	Accurate recommendations	Cold start, sparsity
Demographic-Based	User demographics	Easy to implement	Less personalized
Knowledge-Based	User requirements	No rating history needed	Requires domain knowledge
Utility-Based	Utility function	Flexible	Complex design
Hybrid	Combination of methods	Better accuracy	High complexity

Conclusion

Classical recommendation algorithms provide the foundation for recommendation systems used in social media, e-commerce, streaming services, and online platforms. Among them, **Content-Based Filtering** and **Collaborative Filtering** are the most widely used, while **Hybrid Recommendation Systems** offer improved accuracy by combining the strengths of multiple approaches.

❖ Behaviour Analytics

Apr 2023

Q6) a) Write major components of behavior Analysis Methodology.

Ans. Major Components of Behavior Analysis Methodology

Behavior Analysis Methodology is used in social computing and data mining to understand, predict, and analyze user behavior from social media and online interactions.

The major components are:

1. Data Collection

- Gathering data from various sources such as social media platforms, websites, sensors, and online communities.
- Data may include posts, comments, likes, shares, clicks, and user profiles.

Example: Collecting tweets from users on a social networking site.

2. Data Preprocessing

- Cleaning and transforming raw data into a usable format.
- Removes noise, duplicates, missing values, and irrelevant information.

Activities:

- Data cleaning
- Data integration
- Data transformation
- Feature extraction

3. User Profiling

- Creating a profile that represents user characteristics, preferences, interests, and behavior patterns.
- Profiles may be static or dynamic.

Example: Identifying a user's interest in sports, technology, or entertainment.

4. Behavior Modeling

- Developing models that describe and represent user behavior.
- Uses statistical, machine learning, or graph-based techniques.

Purpose:

- Understand behavior patterns.
- Predict future actions.

5. Pattern Discovery

- Identifying hidden patterns, trends, and relationships in user activities.
- Uses data mining techniques such as:
 - Classification

- Clustering
- Association Rule Mining

Example: Discovering that users who follow technology pages often follow science pages too.

6. Social Network Analysis

- Examining relationships and interactions among users.
- Studies network properties such as:
 - Centrality
 - Communities
 - Influence
 - Connectivity

Example: Finding influential users in a social network.

7. Prediction and Recommendation

- Predicting future user behavior and generating recommendations.
- Uses historical behavior and social connections.

Applications:

- Friend recommendation
- Product recommendation
- Content recommendation

8. Evaluation and Validation

- Measuring the effectiveness and accuracy of behavior analysis models.
- Common metrics include:
 - Accuracy
 - Precision
 - Recall
 - F1-score

Diagram of Behavior Analysis Methodology

Data Collection



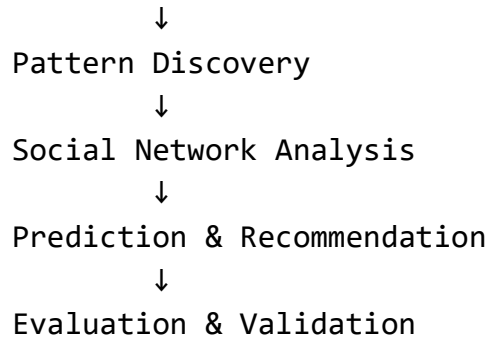
Data Preprocessing



User Profiling



Behavior Modeling



Conclusion

The major components of Behavior Analysis Methodology are **Data Collection, Data Preprocessing, User Profiling, Behavior Modeling, Pattern Discovery, Social Network Analysis, Prediction & Recommendation, and Evaluation & Validation**. Together, these components help analyze user activities, understand behavior patterns, and support decision-making in social computing applications.

❖ Individual Behaviour

Apr 2023

Q5) a) What is Individual Behavior? Explain individual online behavior three categories.

Ans. Individual Behavior

Individual Behavior refers to the actions, reactions, decisions, and activities performed by a single person in response to internal and external factors. In social computing, individual behavior focuses on how a user interacts with online platforms, digital content, and other users.

Understanding individual behavior helps in user profiling, recommendation systems, targeted advertising, and behavior prediction.

Three Categories of Individual Online Behavior

1. Consumption Behavior (Information Consumption)

This behavior involves users **viewing, reading, listening to, or consuming online content** without actively contributing new content.

Examples

- Reading posts on social media.
- Watching videos on YouTube.

- Browsing websites.
- Reading news articles.
- Viewing product descriptions.

Characteristics

- Passive participation.
- Generates browsing and viewing data.
- Helps identify user interests and preferences.

2. Participation Behavior (Interaction Behavior)

This behavior involves users **interacting with existing content or other users** without creating substantial new content.

Examples

- Liking a Facebook post.
- Commenting on a photo.
- Sharing a video.
- Rating a product.
- Following a user or page.

Characteristics

- Active engagement.
- Reflects user opinions and preferences.
- Creates social connections and interactions.

3. Production Behavior (Content Creation)

This behavior involves users **creating and publishing original content** on online platforms.

Examples

- Posting tweets or status updates.
- Uploading videos on YouTube.
- Writing blogs.
- Posting photos on Instagram.
- Creating reviews and articles.

Characteristics

- Highest level of user involvement.
- Generates new information and knowledge.

- Influences other users in the network.

Summary Table

Category	Description	Examples
Consumption Behavior	Viewing and consuming content	Reading posts, watching videos
Participation Behavior	Interacting with content or users	Likes, comments, shares, ratings
Production Behavior	Creating and publishing content	Blogs, videos, posts, reviews

Conclusion

Individual behavior refers to the actions of a single user in an online environment. The three main categories of online behavior are **Consumption Behavior** (using content), **Participation Behavior** (interacting with content), and **Production Behavior** (creating content). These behaviors help researchers and organizations understand user interests, engagement levels, and online activities.

Oct 2023

Q5) b) What is individual behaviour modelling? Explain social community structure.

Ans. Individual Behaviour Modelling

Individual Behaviour Modelling is the process of analyzing, understanding, and predicting the actions, preferences, interests, and decision-making patterns of an individual user based on their past activities and interactions.

It is widely used in **social computing, social media analytics, recommendation systems, marketing, and user profiling**.

Objectives

- Understand user interests and preferences.
- Predict future behavior.
- Personalize content and recommendations.
- Improve user experience.
- Support targeted advertising.

Process of Individual Behaviour Modelling

1. Data Collection

- a. Gather user activities such as clicks, likes, comments, searches, and purchases.
- 2. Feature Extraction**
 - a. Identify relevant behavioral characteristics from collected data.
- 3. User Profiling**
 - a. Create a profile representing user interests and habits.
- 4. Behaviour Analysis**
 - a. Analyze patterns and trends in user activities.
- 5. Prediction**
 - a. Predict future actions and preferences using statistical or machine learning models.

Applications

- Product recommendation systems.
- Friend suggestions in social networks.
- Personalized advertisements.
- Customer behavior analysis.
- Online learning platforms.

Social Community Structure

A **Social Community Structure** refers to the organization of users into groups or communities within a social network where members are more closely connected to each other than to users outside the group.

These communities are formed based on common interests, relationships, activities, professions, locations, or goals.

Definition

A social community is a subset of users in a network that has:

- Strong internal connections.
- Frequent interactions among members.
- Shared interests or characteristics.

Characteristics of Social Communities

1. Dense Connectivity

Members of the same community have many connections with one another.

2. Common Interests

Users often share similar hobbies, professions, opinions, or goals.

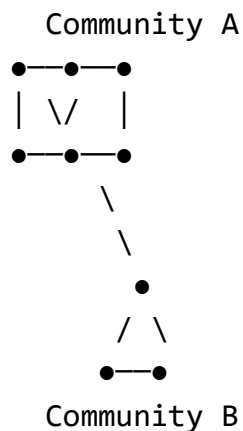
3. Frequent Interaction

Community members communicate and interact regularly.

4. Distinct Boundaries

Communities are more strongly connected internally than externally.

Structure of a Social Community



- Nodes represent users.
- Edges represent relationships or interactions.
- Users within a community are densely connected.

Types of Social Communities

1. Interest-Based Communities

Formed around shared interests.

Example: Photography groups on social media.

2. Professional Communities

Formed based on profession or occupation.

Example: Software developer groups.

3. Location-Based Communities

Formed by users from the same geographic region.

Example: City-specific online communities.

4. Activity-Based Communities

Formed through similar online activities.

Example: Gaming communities.

Community Detection Techniques

- Graph Partitioning
- Hierarchical Clustering
- Girvan-Newman Algorithm
- Modularity-Based Methods
- Spectral Clustering

Applications of Social Community Structure

- Recommendation systems.
- Viral marketing.
- Information diffusion analysis.
- Friend suggestion systems.
- Influence analysis.
- Targeted advertising.

Conclusion

Individual Behaviour Modelling focuses on understanding and predicting the actions of a single user using behavioral data. **Social Community Structure** refers to the grouping of users into closely connected communities based on common interests, interactions, or relationships. Both concepts are fundamental in social computing and social network analysis for improving personalization, recommendations, and understanding user behavior.

❖ Collective Behaviour

Apr 2023

Q5) b) What is collective behavior analysis? Explain user migration in social media.

Ans. Collective Behavior Analysis

Collective Behavior Analysis is the study of the behavior, interactions, and activities of a group of individuals within a social network or online community. Instead of focusing on a single user, it analyzes how groups of users behave collectively and how information, opinions, trends, and actions spread through the network.

Objectives

- Understand group behavior and social dynamics.
- Identify trends and emerging topics.
- Analyze information diffusion.
- Detect communities and influential groups.
- Predict collective actions and social movements.

Characteristics

1. Involves multiple users or communities.
2. Studies interactions among users.
3. Focuses on group patterns rather than individual actions.
4. Uses large-scale social media data.
5. Helps understand social influence and opinion formation.

Examples

- Viral spread of a hashtag on social media.
- Online campaigns and social movements.
- Product adoption by a group of users.
- Community discussions around a trending event.

Applications

- Marketing and advertising.
- Recommendation systems.
- Community detection.
- Trend prediction.
- Public opinion analysis.

User Migration in Social Media

Definition

User Migration refers to the movement of users from one social media platform, community, or online service to another over time. Users may leave an existing platform and join a new one due to changing preferences, better features, privacy concerns, or social influence.

Reasons for User Migration

1. Better Features

Users move to platforms offering improved functionality and user experience.

Example: Moving from traditional social networks to short-video platforms.

2. Social Influence

Friends, family, or influencers encourage users to join another platform.

3. Privacy and Security Concerns

Users may leave platforms due to data privacy issues or security breaches.

4. Content Availability

Users migrate to platforms that provide more relevant or engaging content.

5. Technological Changes

New technologies and innovations attract users to newer platforms.

6. Community Migration

Entire groups or communities may move together to another platform.

Types of User Migration

1. Platform Migration

Movement from one social networking site to another.

Example: Users shifting from one messaging app to another.

2. Community Migration

Users move from one online community to another while maintaining similar interests.

3. Partial Migration

Users remain active on the original platform while also joining a new one.

4. Complete Migration

Users completely abandon one platform and move to another.

Process of User Migration

User Dissatisfaction



Search for Alternatives



Influence from Friends/Community



Join New Platform



Reduce Activity on Old Platform



Migration Completed

Factors Affecting User Migration

- Network effects
- Quality of service
- User experience
- Privacy policies
- Availability of friends and communities
- Content quality
- Platform reputation

Importance of Studying User Migration

1. Helps social media companies retain users.
2. Assists in predicting platform growth or decline.
3. Supports targeted marketing strategies.
4. Helps understand community evolution.
5. Useful for recommendation and engagement analysis.

Conclusion

Collective Behavior Analysis examines the actions and interactions of groups of users to understand social patterns, trends, and information diffusion. **User Migration in Social Media** refers to the movement of users between platforms or communities due to factors such as social influence, better features, privacy concerns, and content preferences. Studying user migration helps organizations understand user retention, platform competition, and social network evolution.

Oct 2023

Q6) a) What is Collective behaviour modelling? Differentiate it with Individual Behaviour modelling.

Ans. Collective Behaviour Modelling

Collective Behaviour Modelling is the process of analyzing, representing, and predicting the behavior of a group of users, communities, or an entire social network rather than a single individual. It studies how people interact, influence one another, form communities, and collectively contribute to trends, information diffusion, and social movements.

The goal is to understand group dynamics and predict large-scale social behaviors.

Objectives

- Analyze group interactions and social dynamics.
- Identify communities and influential groups.
- Study information propagation and opinion formation.
- Predict collective actions and trends.
- Understand social influence and user migration.

Process of Collective Behaviour Modelling

1. **Data Collection** – Gather data from multiple users and communities.
2. **Network Construction** – Create social graphs representing relationships.
3. **Community Detection** – Identify groups of closely connected users.
4. **Pattern Analysis** – Analyze interaction and communication patterns.
5. **Behaviour Prediction** – Predict trends, diffusion, and group activities.

Examples

- Spread of viral content on social media.
- Formation of online communities.
- Adoption of a new product by a user group.
- Public reaction to an event or campaign.

Applications

- Social media analytics.
- Viral marketing.
- Trend prediction.
- Recommendation systems.
- Opinion mining and sentiment analysis.

Difference Between Individual Behaviour Modelling and Collective Behaviour Modelling

Aspect	Individual Behaviour Modelling	Collective Behaviour Modelling	
Definition	Studies and predicts the behavior of a single user.	Studies and predicts the behavior of a group or community.	
Focus	Individual actions, preferences, and interests.	Group interactions, trends, and social dynamics.	
Unit of Analysis	One user.	Multiple users or communities.	
Data Used	User-specific data such as clicks, likes, searches, and purchases.	Data from many users, interactions, and network structures.	
Goal	Understand and predict personal behavior.	Understand and predict collective behavior and trends.	
Relationships Considered	Limited focus on social relationships.	Strong focus on social connections and influence.	
Examples	Recommending movies to a user.	Predicting viral spread of a hashtag.	
Techniques Used	User profiling, classification, recommendation algorithms.	Community detection, social network analysis, diffusion models.	
Output	Personalized recommendations and predictions.	Community trends, influence patterns, and group predictions.	
Application Areas	Personalized advertising, recommendation systems.	Trend analysis, viral marketing, social movement analysis.	

Conclusion

Individual Behaviour Modelling focuses on understanding and predicting the actions of a single user based on personal data and preferences. **Collective Behaviour Modelling** focuses on understanding the behavior of groups, communities, and social networks by analyzing interactions and social influence. While individual modelling supports personalization, collective modelling helps explain large-scale social phenomena such as trends, community formation, and information diffusion.